

Digital Egypt Pioneers Initiative (DEPI)

Graduation Project — Deliverable File 1

PROJECT PLANNING & MANAGEMENT

Houra Platform

Includes: Project Proposal | Project Plan | Task Assignment & Roles | Risk Assessment & Mitigation Plan | KPIs

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Table of Contents

Table of Contents 2

1.1 Executive Summary 3

1.2 Introduction & Background 3

1.3 Problem Statement..... 3

1.4 Proposed Solution – Project Overview 3

1.5 Project Objectives..... 4

1.6 Project Scope..... 4

1.7 Target Users & Stakeholders 4

1.8 Key Features 5

1.9 Business Model Overview 5

1.10 Expected Benefits 5

2.1 Project Timeline (Gantt Chart) 6

2.2 Milestones & Deliverables 6

2.3 Resource Allocation 6

Conclusion..... 10

PART 1 — PROJECT PROPOSAL

1.1 Executive Summary

Time Banking Platform is a digital application that enables individuals to exchange skills and services using time, rather than money, as the unit of value. Every member who provides one hour of service earns one time-credit, which can later be redeemed to receive one hour of service from any other member of the community, regardless of the type of skill involved. The platform directly addresses two connected problems: individuals who possess valuable skills but lack financial resources, and individuals who wish to learn new skills or access services without incurring monetary cost.

This document is the first official deliverable of the graduation project and combines the Project Proposal, Project Plan, Task Assignment & Roles, Risk Assessment & Mitigation Plan, and Key Performance Indicators (KPIs), as required for submission.

1.2 Introduction & Background

Traditional service economies rely on monetary exchange, which inherently excludes individuals who have valuable time and skills but limited financial means, such as students, freelancers, homemakers, and retirees. Time banking is an alternative economic model in which time itself becomes the currency: one hour of any service is always worth one hour of any other service.

The team evaluated two candidate ideas for the graduation project: a “Time Banking Platform” and a “Student Activities Application.” After a structured evaluation using a weighted decision matrix scoring both ideas against cost, creativity, scalability, and value added, the Time Banking Platform scored higher (62 vs. 56) and was selected by team vote as the final project idea.

1.3 Problem Statement

Many individuals possess practical skills and free time but lack the financial resources to access paid services or training. Conversely, many people are willing to share their expertise but have no structured platform through which to do so in a fair, trackable, and trust-based manner. There is a need for a digital platform that:

- Enables fair, non-monetary exchange of skills and services between community members.
- Provides a transparent and trustworthy way to track time contributed and time consumed.
- Encourages collaboration, learning, and community solidarity rather than competition.
- Remains accessible to users regardless of income level.

1.4 Proposed Solution – Project Overview

The Time Banking Platform is a web/mobile digital application where users create profiles listing the skills they can offer and the services they need. When a member provides a service, the hours spent are credited to their time balance; when they receive a service, the equivalent hours are deducted. A rating and review system

builds trust between members, while notifications, search, and filtering tools help members find relevant matches quickly. The platform will start as a free pilot within a defined community before scaling geographically and across service categories.

1.5 Project Objectives

General Objective

To design, develop, and deliver a functional digital Time Banking Platform that allows users to exchange skills and services fairly using time as the unit of value.

Specific Objectives

- Analyze user needs and define clear functional and non-functional requirements.
- Design a reliable database and system architecture capable of tracking time-based transactions accurately.
- Build core features: user profiles, time tracking, ratings and reviews, search, and notifications.
- Implement a secure and trustworthy environment to prevent system exploitation and abuse.
- Test the platform thoroughly to ensure functionality, security, and usability.
- Deliver complete technical and user documentation and a final project presentation.

1.6 Project Scope

In Scope

- User registration, authentication, and profile management.
- Time tracking system (earning and spending time-credits).
- Service listing, search, and advanced filtering.
- Rating and review system, notifications and alerts, in-app messaging.
- Admin dashboard for monitoring and moderation.

Out of Scope (Future Phases)

- Blockchain-based transaction security (long-term enhancement).
- Group-based skill exchange and AI-driven user behavior analysis.
- Integration with external platforms and payment gateways for premium/paid features.
- Multi-city and international expansion.

1.7 Target Users & Stakeholders

- Students and university graduates seeking to build skills and experience.
- Freelancers and independent professionals looking to exchange expertise.
- Homemakers and stay-at-home parents with valuable time and skills.

- Elderly and retired individuals with knowledge to share.
- Non-profit organizations and community initiatives.

1.8 Key Features

Core Features	Advanced Features
User profiles	Advanced filtering & search
Time tracking system	Rewards & incentives
Rating & review system	In-app messaging
Notifications & alerts	Integration with other platforms
—	Group-based skill exchange
—	User behavior analysis

1.9 Business Model Overview

While the platform's core exchange service remains free to preserve accessibility, sustainability will be achieved through a mix of complementary revenue streams:

- Premium subscriptions offering advanced features.
- In-app advertising for small businesses and service providers.
- Small transaction fees applied after an initial number of free exchanges.
- Partnerships with educational institutions and paid workshops or events.

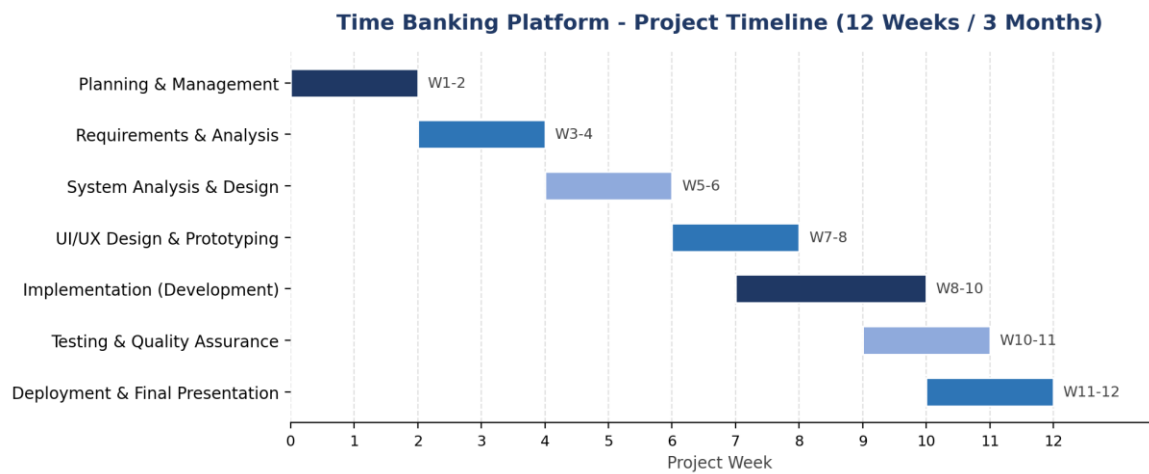
1.10 Expected Benefits

- Promotes collaboration and community solidarity.
- Enables learning new skills without financial cost.
- Supports individuals with valuable skills but limited financial resources.
- Encourages a fair, transparent exchange of knowledge and expertise.

PART 2 — PROJECT PLAN

2.1 Project Timeline (Gantt Chart)

The project will be delivered over a 12-week (3-month) period. The chart below shows the seven major phases and their scheduling across the timeline.



2.2 Milestones & Deliverables

ID	Milestone / Deliverable	Target Date
M1	Project Proposal & Plan Approved	End of Week 2
M2	Requirements Documented & Signed Off	End of Week 4
M3	System Design (ERD, DFD, Architecture) Completed	End of Week 6
M4	UI/UX Wireframes & Mockups Approved	End of Week 8
M5	Core Features Implemented (Working Prototype)	End of Week 10
M6	Testing Completed & Bugs Resolved	End of Week 11
M7	Final Deployment, Documentation & Presentation	End of Week 12

2.3 Resource Allocation

Resource Type	Details
Human Resources	5 team members covering development, design, documentation, and testing.
Development Tools	Code editor (VS Code), Git/GitHub for version control, project management board (Trello/Jira).

Resource Type	Details
Design Tools	Figma or Adobe XD for wireframes and UI/UX mockups.
Backend & Database	Application server and database engine (to be finalized in the Technology Stack section).
Hosting / Deployment	Cloud hosting or university lab servers for the pilot deployment.
Communication	WhatsApp / Discord for daily coordination, weekly meetings with the supervisor.

PART 3 — TASK ASSIGNMENT & ROLES

The table below defines the responsibilities of each of the five team members. Please complete the Name, Role, and Key Responsibilities columns for each member.

#	Team Member Name	Role	Key Responsibilities
1	Amany Mohamed Ebrahim Belal	UI/UX Designer & Frontend Developer & Documentation Lead	Designed the UI/UX wireframes and mockups; implemented the frontend interface of the platform; prepared the technical and user documentation.
2	Nourhan Taha Abdelmonem Mohamed	Backend Developer	Designed and implemented the entire backend system, including the database structure, server-side logic, and API integration.
3	Ahmed Alaa Abdelsalam Ali	Presentation & Coordination	Prepared the final project presentation slides and coordinated meeting schedules with the team and supervisor.
4	Hussein Rezk Abdelrazak Elgorany	Documentation Support & Proofreading	Reviewed and formatted project documents; collected and organized references and sources.
5	Nourhan Moustafa Kamel Mohamed	Testing & Feedback Support	Performed basic manual testing of the platform features and reported feedback/issues to the development team.

Suggested Role Categories (for reference)

- Project Manager / Team Leader – coordinates schedule, meetings, and final deliverables.
- Business Analyst – requirements gathering, use cases, and documentation.
- UI/UX Designer – wireframes, mockups, and design guidelines.
- Backend / Database Developer – server-side logic, database, and API.
- Frontend Developer & QA/Tester – user interface implementation, testing, and bug tracking.

PART 4 — RISK ASSESSMENT & MITIGATION PLAN

The following risks were identified across technical, operational, financial, and adoption dimensions, each rated by likelihood and impact, with a corresponding mitigation strategy.

ID	Risk Description	Category	Likelihood	Impact	Mitigation Strategy
R1	Lack of trust among users toward exchanging services with strangers	Adoption	Medium	High	Implement a strict rating & review system and identity verification at sign-up.
R2	Imbalance between supply and demand for certain skills	Operational	Medium	Medium	Offer rewards/incentives for in-demand services; promote underrepresented categories.
R3	System exploitation (users gaming the time-credit system)	Technical / Security	Medium	High	Set a minimum required service contribution before requesting services; monitor unusual activity.
R4	Difficulty generating revenue in the early stage	Financial	High	Medium	Start with a free pilot phase before introducing paid/premium features.
R5	Delay in development due to limited team experience with new tools	Schedule	Medium	Medium	Allocate buffer time in the plan; conduct short technical training sessions early on.
R6	Server/infrastructure costs exceeding budget as user base grows	Technical / Financial	Low	High	Start with low-cost cloud hosting tiers; scale infrastructure gradually with usage.
R7	Low user engagement after launch	Adoption	Medium	High	Run awareness campaigns, onboarding incentives, and a “first hour free” program.
R8	Data privacy and security concerns	Security	Low	High	Apply secure authentication, data encryption, and clear privacy policies.

PART 5 — KEY PERFORMANCE INDICATORS (KPIs)

The following KPIs will be used to measure the success of the Time Banking Platform, both technically and in terms of user adoption.

KPI	Target	Measurement Method
System Response Time	≤ 2 seconds for 95% of requests	Server-side performance monitoring / load testing
System Uptime	≥ 99% availability	Hosting/monitoring dashboard (e.g., UptimeRobot)
User Adoption Rate	300+ registered users within first 2 months of pilot	Platform analytics / admin dashboard
Monthly Active Exchanges	150+ completed service exchanges per month	Transaction logs
User Satisfaction	≥ 85% positive rating (4/5 stars or higher)	In-app rating & review system, surveys
Bug Resolution Rate	≥ 90% of reported bugs resolved before final submission	Bug tracking sheet / QA reports
Task/Sprint Completion Rate	≥ 90% of planned tasks completed on schedule	Project management board (Trello/Jira)

Conclusion

This document constitutes the complete first deliverable of the graduation project, covering the Project Proposal, Project Plan, Task Assignment & Roles, Risk Assessment & Mitigation Plan, and Key Performance Indicators. Together, these establish a clear roadmap and accountability framework for the successful delivery of the Time Banking Platform within the three-month project timeline.